

VIX Regimes and Equity Market Behavior

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1 Overview

This report examines how volatility regimes influence forward equity returns and realized volatility. Using SPY and VIX data, the market is segmented into four regimes and analyzed across multiple forward horizons.

Rather than treating volatility as a passive measure of risk, this analysis frames it as an active driver of return dynamics and market behavior.

2 VIX Regime Definition

Regime	Definition
Low Vol	$VIX < 15$
Normal Vol	15–20
Elevated Vol	20–30
Stress Vol	> 30

These regimes capture distinct market environments, ranging from stable, low-risk periods to stressed conditions characterized by uncertainty and dislocation.

3 Market Context

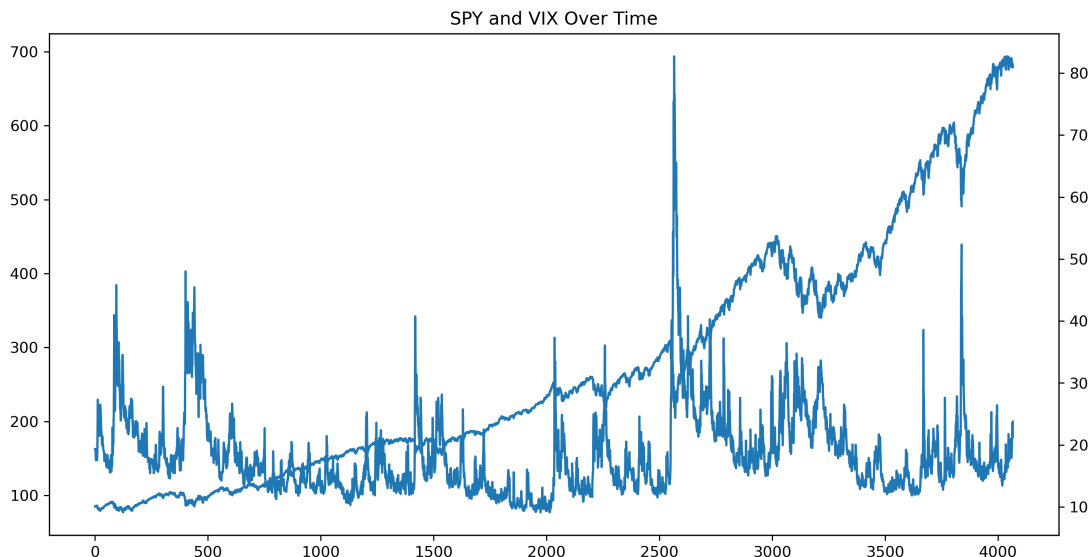


Figure 1: SPY and VIX Over Time

The relationship between SPY and VIX reflects a well-known dynamic: volatility spikes tend to coincide with market drawdowns. However, these periods also set the stage for future returns.

4 Forward Returns by Regime

Table 1: Average Forward Returns			
Regime	1D	5D	20D
Elevated	0.000675	0.004149	0.015760
Low	0.000406	0.001432	0.006569
Normal	0.000332	0.002072	0.008317
Stress	0.002474	0.010292	0.039222

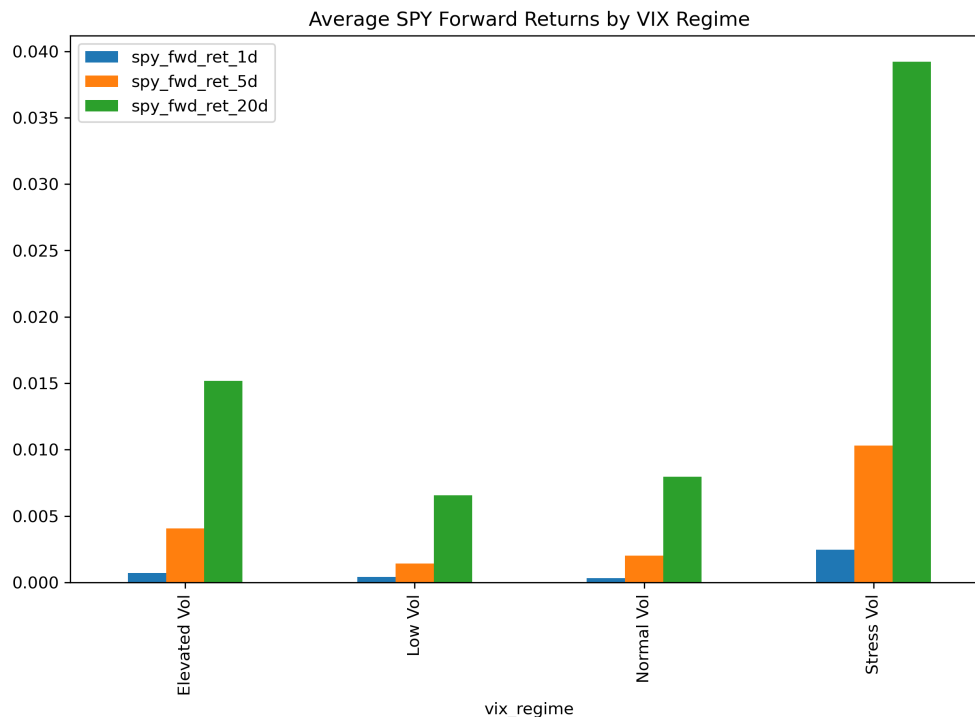
Key Insight: Forward returns increase meaningfully as volatility rises, with the strongest effect observed in stress regimes.

Explanation: During high-volatility environments, markets are typically undergoing drawdowns, deleveraging, and liquidity stress. These conditions push prices below fundamental value, creating elevated expected returns going forward.

Interpretation:

- Low-vol regimes reflect stable, fully-priced markets with limited upside
- Elevated and stress regimes reflect dislocations and forced selling
- The stronger effect at 20D horizons suggests mean reversion dominates over time

Takeaway: Volatility spikes are not just risk events—they are opportunities. Increasing exposure during high-vol regimes improves expected returns, particularly over medium horizons.



5 Risk-Adjusted Returns

Table 2: Sharpe Ratios by Regime

Regime	5D Sharpe	20D Sharpe
Elevated	0.1488	0.3207
Low	0.1041	0.2125
Normal	0.1112	0.2136
Stress	0.2259	0.6317

Insight: Stress regimes deliver the highest risk-adjusted returns.

Explanation: In stressed markets, correlations increase and liquidity deteriorates, forcing participants to unwind positions. This leads to overshooting in prices, which later corrects as conditions normalize.

Takeaway:

- High volatility improves both absolute and risk-adjusted returns
- Volatility should be viewed as a source of alpha, not just risk
- Scaling into drawdowns is more effective than static allocation

6 Future Realized Volatility

Table 3: Future Realized Volatility

Regime	5D Vol	20D Vol
Elevated	0.1867	0.1881
Low	0.0820	0.1037
Normal	0.1214	0.1341
Stress	0.3652	0.3091

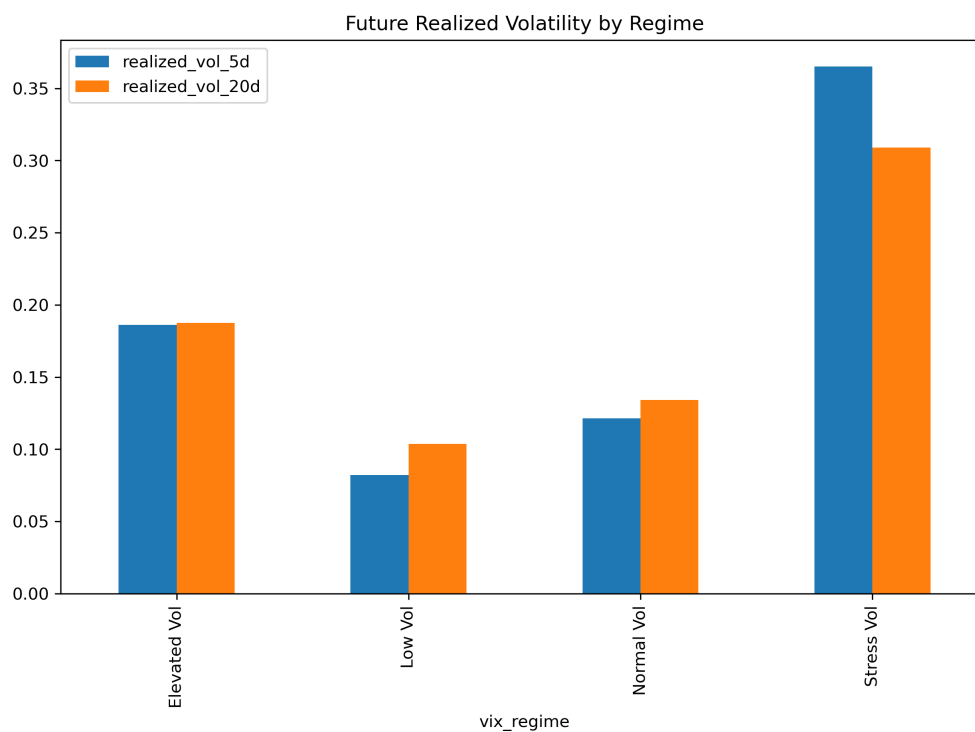
Insight: High VIX strongly predicts elevated future realized volatility.

Explanation: The VIX reflects implied volatility from options markets and captures forward-looking expectations. When elevated, it signals sustained uncertainty rather than a one-off shock.

Interpretation:

- Volatility is persistent in the short term
- High-vol regimes reflect continued instability
- This has direct implications for hedging and options pricing

Takeaway: The VIX is an effective predictor of future volatility and a key input in both risk management and volatility trading.



7 VIX Spike Analysis

Spike Threshold: 0.1362

Table 4: Returns After VIX Spike			
Type	1D	5D	20D
Non-Spike	0.000539	0.002878	0.011277
Spike	0.001188	0.002006	0.012383

t-stat: -0.374 p-value: 0.7088

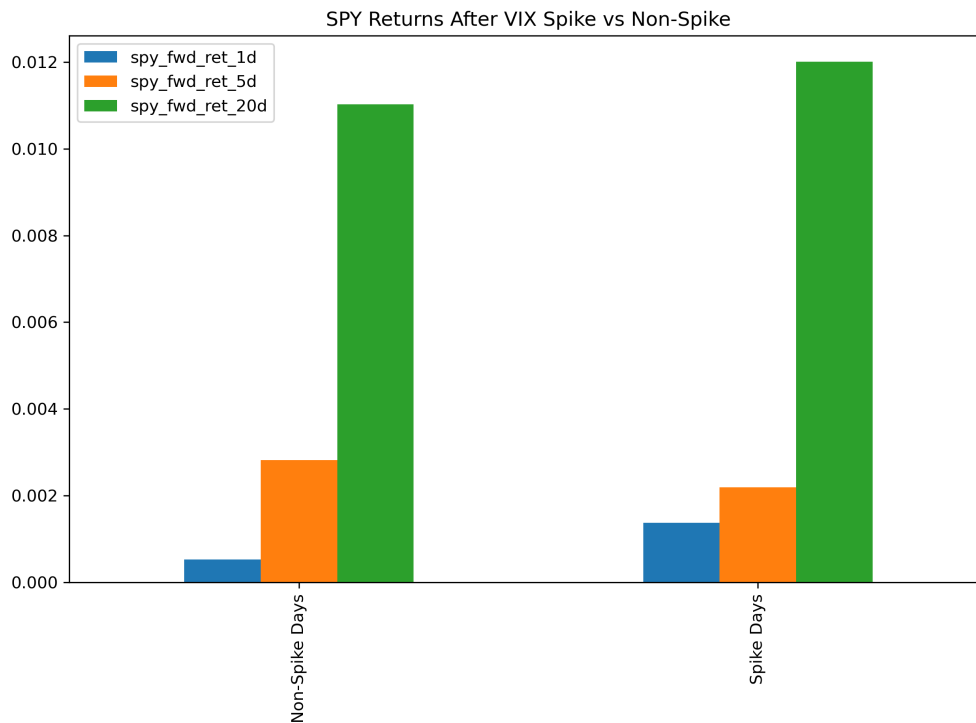
Interpretation: There is no statistically significant difference in returns following single-day volatility spikes.

Explanation: Single-day spikes often reflect short-term news or positioning rather than a structural shift in market conditions.

Key Distinction:

- Spikes are transient and noisy
- Regimes are persistent and informative

Takeaway: Regime-based signals are more reliable than reacting to isolated volatility events.



8 Key Takeaways (EQD Perspective)

- Volatility regimes are a primary driver of return dynamics
- High-volatility environments generate stronger returns and Sharpe ratios
- Implied volatility is a strong predictor of future realized volatility
- Market stress creates pricing dislocations and alpha opportunities

- Persistent regimes matter more than short-term spikes

9 Conclusion

Volatility is not merely a measure of risk—it is a structural feature of markets that shapes returns, risk premia, and trading opportunities.

This analysis shows that high-volatility regimes are associated with both elevated returns and improved risk-adjusted performance, driven by dislocations and mean reversion. At the same time, volatility is persistent, making the VIX a useful forward-looking signal.

From an equity derivatives perspective, incorporating regime-based frameworks provides a more robust and actionable approach than relying on isolated signals.